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SELECTIVELY DEPLOYABLE OTTOMAN FOR A VEHICLE

Abstract

A selectively deployable ottoman includes a base member including an adjustment axis and an ottoman assembly connected to the base member. The ottoman assembly includes a first member having a first end and a second end. The first end being connected to the base member. A second member has a first end portion and a second end portion. The first end portion being pivotally connected to the second end of the first member. A third member includes a first end section and a second end section. The first end section is pivotally connected to the second end portion of the second member. A connector member operatively connects the first end of the first member to the base member. The connector member being shiftable along the adjustment axis of the base member to selectively deploy the ottoman assembly.

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Background/Summary

INTRODUCTION

[0001] The subject disclosure relates to the art of vehicles and, more particularly, to a selectively deployable ottoman for a vehicle.

[0002] Modern vehicles include amenities that would have never entered the imagination of designers of luxury cars of old, let alone to the design of a passenger vehicle slated for wide use. Many modern cars include movie screens, video players, surround sound, reclining seats, and the like. Vehicle manufacturers strive to increase occupant comfort for all vehicles, especially autonomous and semi-autonomous vehicles. As autonomous and semi-autonomous vehicles develop, the need to develop new amenities that further increase occupant comfort increases.

[0003] Sitting for prolonged periods in a vehicle seat has been shown to increase occupant discomfort. Often times, frequent stops are integrated into longer trips. The stops provide passengers an opportunity to get out, stretch, and otherwise change body position. Increasing occupant comfort, particularly leg comfort, may lead to better ergonomics which may reduce the need to stop the vehicle and adjust body position. Increasing leg comfort will therefore reduce unnecessary stops. Accordingly, it is desirable to provide a selectively deployable leg support in vehicle occupant spaces.

SUMMARY

[0004] A selectively deployable ottoman for a vehicle, in accordance with a non-limiting example, includes a base member including an adjustment axis and an ottoman assembly connected to the base member. The ottoman assembly including a first member including a first end and a second end that is opposite the first end. The first end being connected to the base member. A second member has a first end portion and a second end portion that is opposite the first end portion. The first end portion being pivotally connected to the second end of the first member. A third member includes a first end section and a second end section that is opposite the first end section. The first end section is pivotally connected to the second end portion of the second member. A connector member operatively connects the first end of the first member to the base member. The connector member being shiftable along the adjustment axis of the base member to selectively deploy the ottoman assembly.

[0005] In addition to one or more of the features described herein a first pivot axis and a second pivot axis arranged at the second end portion of the third member.

[0006] In addition to one or more of the features described herein the first pivot axis and the second pivot axis comprise an axle that extends through the third member.

[0007] In addition to one or more of the features described herein a torsion spring arranged about one of the first pivot axis and the second pivot axis.

[0008] In addition to one or more of the features described herein a first deployment member operatively connected between the second end of the first member and the second end portion of the third member.

[0009] In addition to one or more of the features described herein a second deployment member operatively connected between the first end of the first member and the second end portion of the second member.

[0010] In addition to one or more of the features described herein the first deployment member comprises a first linear actuator including a first housing and a first piston that is selectively extendable from the first housing and the second deployment member comprises a second linear actuator including a second housing and a second piston that is selectively extendable from the second housing.

[0011] In addition to one or more of the features described herein the first housing is pivotally connected to the second end portion of the third member and the first piston is pivotally connected to the second end of the first member and the second housing is pivotally connected to the second end section of the second member and the first end of the first member.

[0012] In addition to one or more of the features described herein a first guide member and a second guide member secured to the base member, the connector member being operatively connected to the first guide member and the second guide member.

[0013] In addition to one or more of the features described herein the base member is mounted to a floor surface of the vehicle, the base member being selectively shiftable relative to the floor surface along the adjustment axis.

[0014] A vehicle, in accordance with a non-limiting example, includes a body including a passenger compartment having a floor surface and a selectively deployable ottoman arranged in the passenger compartment. The selectively deployable ottoman including a base member connected to the floor surface, the base member including an adjustment axis and an ottoman assembly connected to the base member. The ottoman assembly includes a first member including a first end and a second end that is opposite the first end. The first end being connected to the base member. A second member has a first end portion and a second end portion that is opposite the first end portion. The first end portion is pivotally connected to the second end of the first member. A third member includes a first end section and a second end section that is opposite the first end section. The first end section is pivotally connected to the second end portion of the second member. A connector member operatively connecting the first end of the first member to the base member. The connector member is shiftable along the adjustment axis of the base member to selectively deploy the ottoman assembly.

[0015] In addition to one or more of the features described herein a first pivot axis and a second pivot axis arranged at the second end portion of the third member.

[0016] In addition to one or more of the features described herein the first pivot axis and the second pivot axis comprise an axle that extends through the third member.

[0017] In addition to one or more of the features described herein a torsion spring arranged about one of the first pivot axis and the second pivot axis.

[0018] In addition to one or more of the features described herein a first deployment member operatively connected between the second end of the first member and the second end portion of the third member.

[0019] In addition to one or more of the features described herein a second deployment member operatively connected between the first end of the first member and the second end portion of the second member.

[0020] In addition to one or more of the features described herein the first deployment member comprises a first linear actuator including a first housing and a first piston that is selectively extendable from the first housing and the second deployment member comprises a second linear actuator including a second housing and a second piston that is selectively extendable from the second housing.

[0021] In addition to one or more of the features described herein the first housing is pivotally connected to the second end portion of the third member and the first piston is pivotally connected to the second end of the first member and the second housing is pivotally connected to the second end section of the second member and the first end of the first member.

[0022] In addition to one or more of the features described herein a first guide member and a second guide member secured to the base member, the connector member being operatively connected to the first guide member and the second guide member.

[0023] In addition to one or more of the features described herein the base member is selectively shiftable relative to the floor surface along the adjustment axis.

[0024] The above features and advantages, and other features and advantages of the disclosure are readily apparent from the following detailed description when taken in connection with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] Other features, advantages and details appear, by way of example only, in the following detailed description, the detailed description referring to the drawings in which:

[0026] FIG. 1 is a left side view of a vehicle having a door removed to depict a selectively deployable ottoman in a stowed configuration, in accordance with a non-limiting example;

[0027] FIG. 2 is a left side view of a vehicle having a door removed to depict a selectively deployable ottoman in a calf support configuration, in accordance with a non-limiting example;

[0028] FIG. 3 is a left side view of a vehicle having a door removed to depict a selectively deployable ottoman in a foot rest configuration, in accordance with a non-limiting example;

[0029] FIG. 4 is a plan view of the selectively deployable ottoman in the stowed configuration, in accordance with a non-limiting example;

[0030] FIG. 5 is a side view of the selectively deployable ottoman in a deployed configuration, in accordance with a non-limiting example;

[0031] FIG. 6 is a detail view of first and second torsion springs on the selectively deployable ottoman, in accordance with a non-limiting example; and

[0032] FIG. 7 is a block diagram illustrating a control system for the selectively deployable ottoman, in accordance with a non-limiting example.

DETAILED DESCRIPTION

[0033] The following description is merely exemplary in nature and is not intended to limit the present disclosure, its application or uses. It should be understood that throughout the drawings, corresponding reference numerals indicate like or corresponding parts and features. As used herein, the term module refers to processing circuitry that may include an application specific integrated circuit (ASIC), an electronic circuit, a processor (shared, dedicated, or group) and memory that executes one or more software or firmware programs, a combinational logic circuit, and/or other suitable components that provide the described functionality.

[0034] A vehicle, in accordance with a non-limiting example, is indicated generally at **10** in FIG. 1. Vehicle **10** includes a body **12** supported on a plurality of wheels **16**. Body **12** defines, in part, a passenger compartment **20** having seats **23** positioned behind a dashboard **26**. Passenger compartment **20** is accessed through a door (not shown in order to enhance clarity). A steering control **30** is arranged between seats **23** and dashboard **26**. Steering control **30** is operated to control orientation of select ones of the plurality of wheels **16**.

[0035] Vehicle **10** includes an electric machine shown in the form of an electric drive unit **34** that provides power to one or more of the plurality of wheels **16**. Electric drive unit **34** is operatively connected to a rechargeable energy storage system (RESS) or battery assembly **38**. While shown as being an electric vehicle, it should be understood that vehicle **10** may be a hybrid vehicle or a vehicle powered by an internal combustion engine (ICE). Further, vehicle **10** may be an autonomous vehicle such that a driver need not manipulate controls, (i.e., steering, acceleration, braking etc.) Such controls, including steering, acceleration, braking, and the like could also be collapsed and hidden if and when vehicle **10** is in full autonomous driving mode. It should be understood that vehicle **10** may also take the form of a human operated vehicle.

[0036] In a non-limiting example, passenger compartment **20** includes a passenger compartment floor **40** having a floor surface **42**. Floor surface **42** supports a selectively deployable ottoman **50**. Selectively deployable ottoman **50** may transition from a first or stowed configuration as shown in FIG. 1, to a second or calf support configuration, such as shown in FIG. 2, and to a foot rest configuration, such as shown in FIG. 3. At this point it should be understood that while shown as being associated with a driver's position in passenger compartment **20**, selectively deployable ottoman **50** may be arranged in other locations. The particular location(s) of selectively deployable

ottoman **50** may vary and may depend on vehicle configuration (autonomous, semi-autonomous, human controlled).

[0037] In a non-limiting example shown in FIG. 4, selectively deployable ottoman **50** includes a base member **60** slideably connected to floor surface **42**. Base member **60** is mounted to floor surface **42** through a guide track **62** that may include a motor **64**. As will be detailed more fully herein, motor **64** is selectively controlled to shift base member **60** relative to floor surface **42** along an adjustment axis "A". In a non-limiting example, adjustment axis "A" extends along a longitudinal axis or fore-to-aft in passenger compartment **20**. Base member **60** includes a support surface **69** that supports selectively deployable ottoman **50**.

[0038] Reference will now follow to FIGS. 4 and 5 and with continued reference to FIG. 3 in describing selectively deployable ottoman **50** in accordance with a non-limiting example. Selectively deployable ottoman **50** includes an ottoman assembly **80** including a first member **82**, a second member **84** operatively and pivotally connected to first member **82**, and a third member **86** operatively and pivotally connected to second member **84**. First member **82** is shiftably coupled to support surface **69** through a connector member **90**. In a non-limiting example, connector member **90** transitions over support surface **69** through a first guide member **92** and a second guide member **94**. First and second guide members **92** and **94** may be embedded in base member **60**.

[0039] In a non-limiting example, first member **82** includes a first end **100** and a second end **102** that is opposite first end **100**. First end **100** is separated from second end **102** by a first support surface **104**. Second member **84** includes a first end portion **108** and a second end portion **110** that is opposite first end portion **108**. First end portion **108** is separated from second end portion **110** by a second support or calf support surface **112**. Third member **86** includes a first end section **120** and a second end section **122** that is opposite to first end section **120**. First end section **120** is separated from second end section **122** by a foot rest surface **124**.

[0040] Referring to FIG. 5 and with continued reference to FIG. 4, first end **100** of first member **82** is coupled to connector member **90** through a first hinge **130** that defines a first pivot axis **132**. First hinge **130** may comprise a single hinge that extends along a width of first member **82** or may comprise multiple hinge elements. First end portion **108** of second member **84** is coupled to second end **102** of first member **82** through a second hinge **136** that defines a second pivot axis **138**. In a manner similar to that described with respect to first hinge **130**, second hinge **136** may comprise a single hinge that extends along a width of second member **84** or may comprise multiple hinge elements.

[0041] First end section **120** of third member **86** is coupled to second end portion **110** of second member **84** through a third hinge **142** that defines a third pivot axis **144**. In a manner similar to that described with respect to first hinge **130** and second hinge **136**, third hinge may comprise a single hinge that extends along a width of third member **86** or may comprise multiple hinge elements. A fourth hinge **148** that defines a fourth pivot axis **160** is arranged at second end section **122** of third member **86**. Fourth hinge **148** may be formed from one or more hinge elements.

[0042] In an example shown in FIG. 6 and with continued reference to FIG. 5, second end section **122** includes a first pivot **162** and a second pivot **164**. First pivot **162** and second pivot **164** may be embodied in an axle **168** that defines fourth pivot axis **160**. Axle **168** may define a continuous member that extends across a width of third member **86** or may define discrete axles that embody first pivot **162** and second pivot **164** respectively. Axle **168** may support one or more torsion springs **170** that control deployment of selectively deployable ottoman **50** as will be detailed more fully herein.

[0043] In a non-limiting example, selectively deployable ottoman **50** includes a first deployment member **180** and a second deployment member **182**. First deployment member **180** and second deployment member **182** are selectively activated to transition ottoman assembly **80** between the stowed configuration (FIG. 1) and a deployed configuration (FIG. 2 and FIG. 3). First deployment member **180** may take the form of a first linear actuator **190** and second deployment member **182**

may take the form of a second linear actuator **192**. At this point, it should be understood that first deployment member **180** and second deployment member **182** may take on various forms including systems that rely on non-linear actuation such as rotation, or hybrid systems that may rely on linear and non-linear actuation.

[0044] First linear actuator **190** includes a first housing **200** and a first piston **202**. Second linear actuator **192** includes a second piston **204** and a second housing **206**. First piston **202** includes a first piston end **208** connected to second end **102** of first member **82** and a second piston end (not shown) that extends into first housing **200**. Second piston **204** includes a first piston end portion **211** connected to second end portion **110** of second member **84** and a second piston end portion (not shown) that extends into second housing **206**. First housing **200** includes a first housing end **216** connected to second end section **122** of third member **86** and a second housing end **218** that is receptive of the second piston end of first piston **202**. Similarly, second housing **206** includes a first housing end section **222** connected to first end **100** of first member **82** and a second housing end section **224** that is receptive of the second piston end portion **110**.

[0045] From the stowed configuration, contraction of first piston **202** into first housing **200** and second piston **204** into second housing **206** causes ottoman assembly **80** to rise relative to base member **60**. As ottoman assembly **80** lifts, third member **86** rotates about first pivot **162** and second pivot **164**. Connector member **90** transitions over support surface **69** along adjustment axis “A” in first guide member **92** and second guide member **94**. As third member **86** rotates, torsion spring **170** is loaded so as to control movement speed and promote rotation in a selected direction. At this point, motor **64** may be activated to shift base member **60** along floor surface in order to position ottoman assembly **80** the second or calf support configuration, such as shown in FIG. 2, or the foot rest configuration, such as shown in FIG. 3. Of course, it should be understood that base member **60** may be shifted before ottoman assembly **80** moves, or base member **60** and ottoman assembly **80** may move in unison.

[0046] In a non-limiting example shown in FIG. 7, vehicle **10** includes an ottoman control system **236** operatively connected to motor **64** associated with movement of base member **60** as well as first deployment member **180** and second deployment member **182**. Ottoman control system **236** includes a central processing unit (CPU) **240**, a non-volatile memory **246**, and an ottoman controller **248**. Ottoman control system **236** may also include sensors that record ottoman position. Ottoman position may also simply be temporarily stored in non-volatile memory **246**. Vehicle **10** may include a toggle **250** that selectively controls deployment of ottoman assembly **80**. Ottoman assembly **80** may be deployed to increase passenger comfort on longer rides thereby reducing the need for or frequency of unscheduled stops.

[0047] The terms “a” and “an” do not denote a limitation of quantity, but rather denote the presence of at least one of the referenced item. The term “or” means “and/or” unless clearly indicated otherwise by context. Reference throughout the specification to “an aspect”, means that a particular element (e.g., feature, structure, step, or characteristic) described in connection with the aspect is included in at least one aspect described herein, and may or may not be present in other aspects. In addition, it is to be understood that the described elements may be combined in any suitable manner in the various aspects.

[0048] When an element such as a layer, film, region, or substrate is referred to as being “on” another element, it can be directly on the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly on” another element, there are no intervening elements present.

[0049] Unless specified to the contrary herein, all test standards are the most recent standard in effect as of the filing date of this application, or, if priority is claimed, the filing date of the earliest priority application in which the test standard appears.

[0050] Unless defined otherwise, technical and scientific terms used herein have the same meaning as is commonly understood by one of skill in the art to which this disclosure belongs.

[0051] While the above disclosure has been described with reference to exemplary embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from its scope. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the disclosure without departing from the essential scope thereof. Therefore, it is intended that the present disclosure not be limited to the particular embodiments disclosed, but will include all embodiments falling within the scope thereof.

Claims

1. A selectively deployable ottoman for a vehicle, the selectively deployable ottoman comprising: a base member including an adjustment axis; and an ottoman assembly connected to the base member, the ottoman assembly comprising: a first member including a first end and a second end that is opposite the first end, the first end being connected to the base member; a second member having a first end portion and a second end portion that is opposite the first end portion, the first end portion being pivotally connected to the second end of the first member; a third member including a first end section and a second end section that is opposite the first end section, the first end section being pivotally connected to the second end portion of the second member; and a connector member operatively connecting the first end of the first member to the base member, the connector member being shiftable along the adjustment axis of the base member to selectively deploy the ottoman assembly.
2. The selectively deployable ottoman according to claim 1, further comprising a first pivot axis and a second pivot axis arranged at the second end portion of the third member.
3. The selectively deployable ottoman according to claim 2, wherein the first pivot axis and the second pivot axis comprise an axle that extends through the third member.
4. The selectively deployable ottoman according to claim 3, further comprising a torsion spring arranged about one of the first pivot axis and the second pivot axis.
5. The selectively deployable ottoman according to claim 1, further comprising a first deployment member operatively connected between the second end of the first member and the second end portion of the third member.
6. The selectively deployable ottoman according to claim 5, further comprising a second deployment member operatively connected between the first end of the first member and the second end portion of the second member.
7. The selectively deployable ottoman according to claim 6, wherein the first deployment member comprises a first linear actuator including a first housing and a first piston that is selectively extendable from the first housing and the second deployment member comprises a second linear actuator including a second housing and a second piston that is selectively extendable from the second housing.
8. The selectively deployable ottoman according to claim 7, wherein the first housing is pivotally connected to the second end portion of the third member and the first piston is pivotally connected to the second end of the first member and the second housing is pivotally connected to the second end section of the second member and the first end of the first member.
9. The selectively deployable ottoman according to claim 8, further comprising a first guide member and a second guide member secured to the base member, the connector member being operatively connected to the first guide member and the second guide member.
10. The selectively deployable ottoman according to claim 1, wherein the base member is mounted to a floor surface of the vehicle, the base member being selectively shiftable relative to the floor surface along the adjustment axis.
11. A vehicle comprising: a body including a passenger compartment having a floor surface; and a selectively deployable ottoman arranged in the passenger compartment, the selectively deployable

ottoman comprising: a base member connected to the floor surface, the base member including an adjustment axis; and an ottoman assembly connected to the base member, the ottoman assembly comprising: a first member including a first end and a second end that is opposite the first end, the first end being connected to the base member; a second member having a first end portion and a second end portion that is opposite the first end portion, the first end portion being pivotally connected to the second end of the first member; a third member including a first end section and a second end section that is opposite the first end section, the first end section being pivotally connected to the second end portion of the second member; and a connector member operatively connecting the first end of the first member to the base member, the connector member being shiftable along the adjustment axis of the base member to selectively deploy the ottoman assembly.

12. The selectively deployable ottoman according to claim 11, further comprising a first pivot axis and a second pivot axis arranged at the second end portion of the third member.

13. The selectively deployable ottoman according to claim 12, wherein the first pivot axis and the second pivot axis comprise an axle that extends through the third member.

14. The selectively deployable ottoman according to claim 13, further comprising a torsion spring arranged about one of the first pivot axis and the second pivot axis.

15. The selectively deployable ottoman according to claim 11, further comprising a first deployment member operatively connected between the second end of the first member and the second end portion of the third member.

16. The selectively deployable ottoman according to claim 15, further comprising a second deployment member operatively connected between the first end of the first member and the second end portion of the second member.

17. The selectively deployable ottoman according to claim 16, wherein the first deployment member comprises a first linear actuator including a first housing and a first piston that is selectively extendable from the first housing and the second deployment member comprises a second linear actuator including a second housing and a second piston that is selectively extendable from the second housing.

18. The selectively deployable ottoman according to claim 17, wherein the first housing is pivotally connected to the second end portion of the third member and the first piston is pivotally connected to the second end of the first member and the second housing is pivotally connected to the second end section of the second member and the first end of the first member.

19. The selectively deployable ottoman according to claim 18, further comprising a first guide member and a second guide member secured to the base member, the connector member being operatively connected to the first guide member and the second guide member.

20. The selectively deployable ottoman according to claim 11, wherein the base member is selectively shiftable relative to the floor surface along the adjustment axis.
